

澳門理工大學
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COMP2113 OPERATING SYSTEMS

File Management

(Ref: Chapter 12)

We'll learn in this chapter...

- File , and directory
- File systems
- Access control – Users or groups of users are granted certain access rights to a file
- Simultaneous access – file locking
- Secondary Storage Management, File system:
  - FAT, NTFS

Files

- Data collections created by users
- The File System is one of the most important parts of the OS to a user
- Desirable properties of files:

Long-term existence

- files are stored on disk or other secondary storage and do not disappear when a user logs off

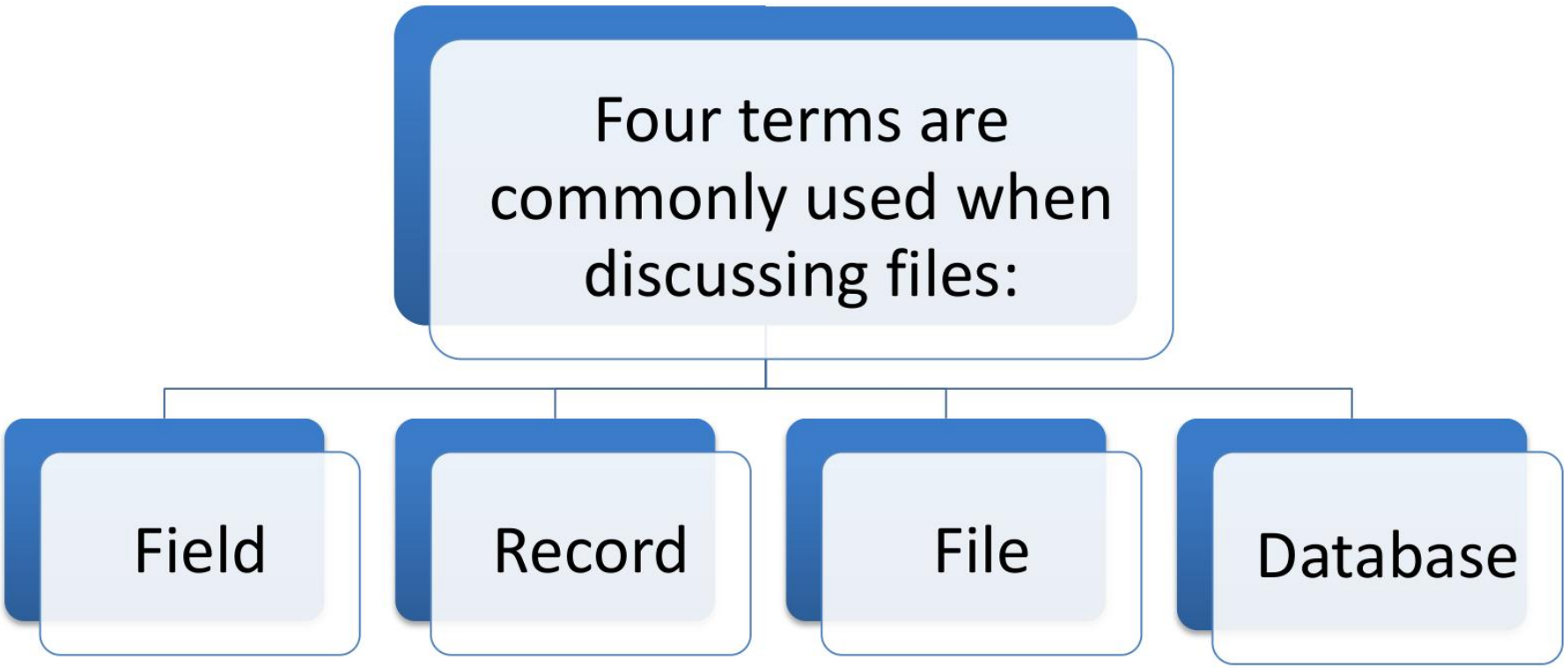
Sharable between processes

- files have names and can have associated access permissions that permit controlled sharing

Structure

- files can be organized into hierarchical or more complex structure to reflect the relationships among files

File Structure



File Directories

- Contains information about files
  - attributes: read only, creation date, size, etc..
  - location: where the file resides on hard disk
  - ownership
- Directory itself is a file

Access Rights

- None
  - user may not know the existence of the file
- Knowledge
  - user can only determine that the file exists and who its owner is
- Execution
  - can load and execute a program but cannot copy it

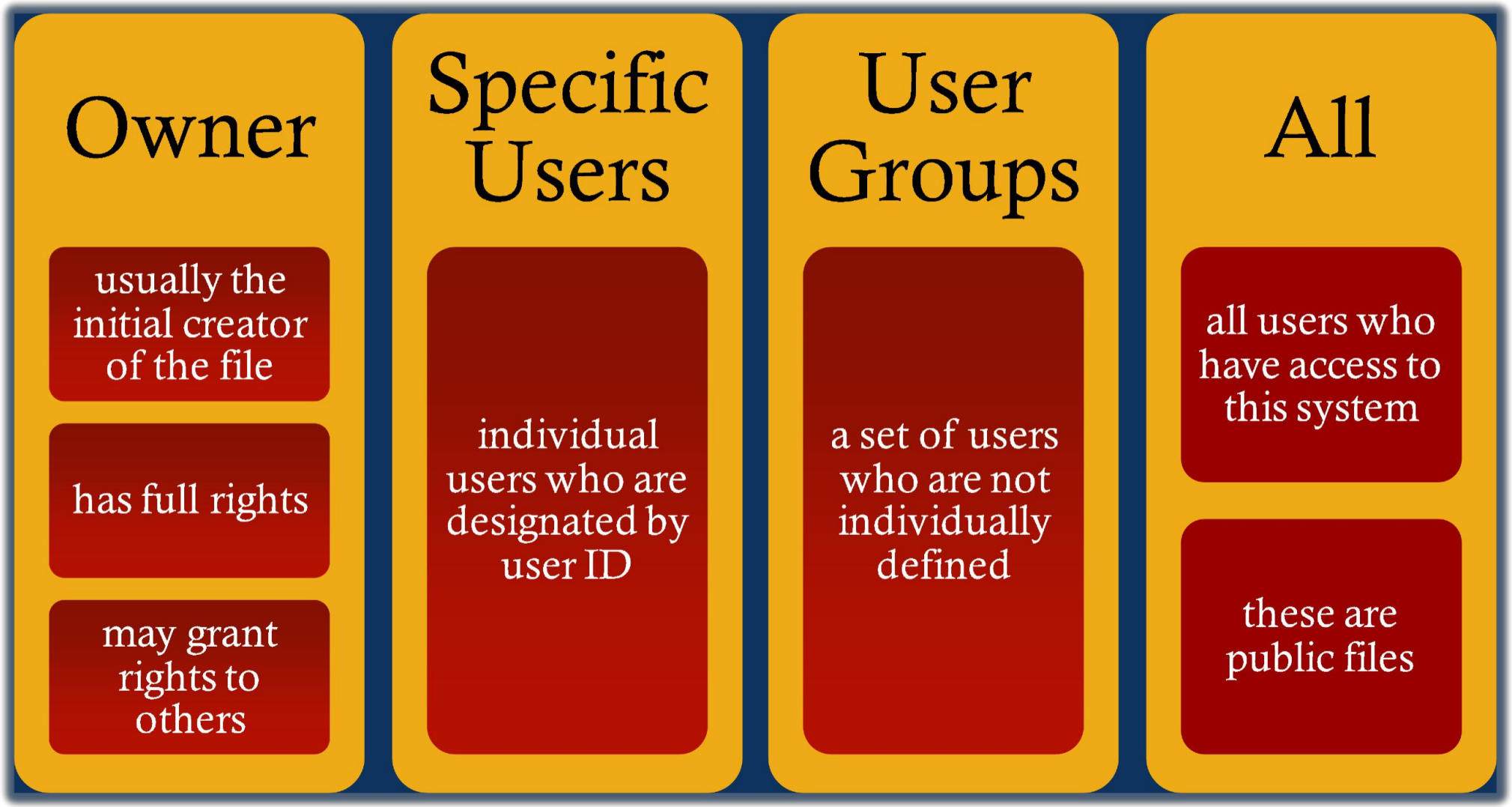
Access Rights

- Reading
  - can read the file (incl. copying and execution)
- Appending
  - can only add data to the file
- Updating
  - can modify, delete, and add data
- Changing protection
  - can change access rights granted to other users

Access Rights

- Deletion
  - can delete the file
- Owners
  - has all rights previously listed
  - may grant rights to others

User Access Rights



Simultaneous Access

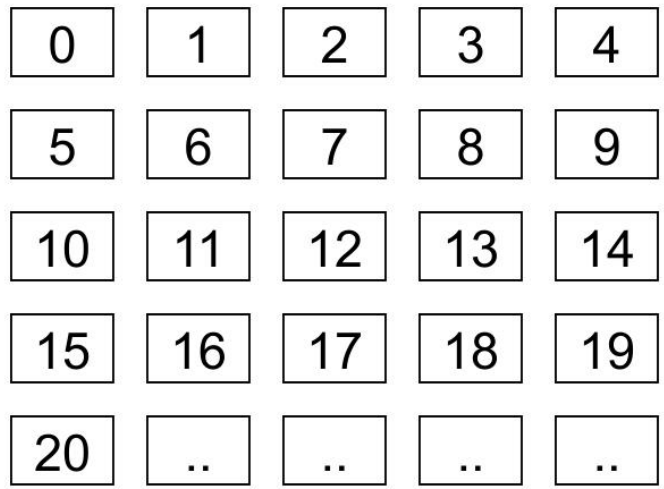
- Readers/Writers problem
- User may lock entire file before updating
- User may lock the individual records during the update

Secondary Storage Management

- Space on disks is allocated to each file
- File system must keep track of which parts are allocated to which file, and also the available space
- Space is allocated as one or more contiguous units or portions

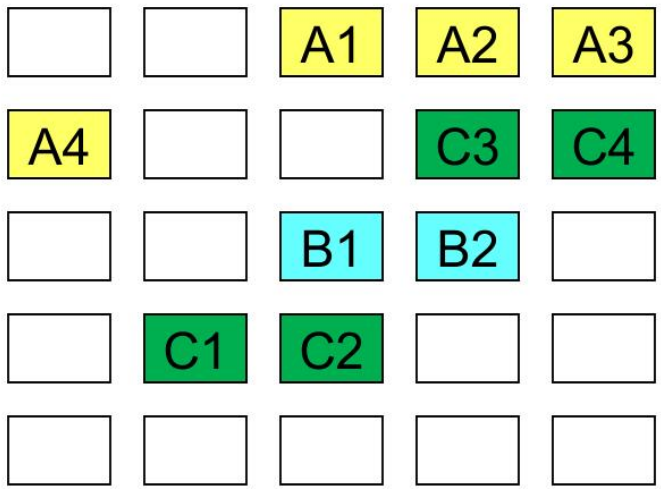
Cluster

- For the OS, the hard disk storage space is divided in equal size chunks called cluster



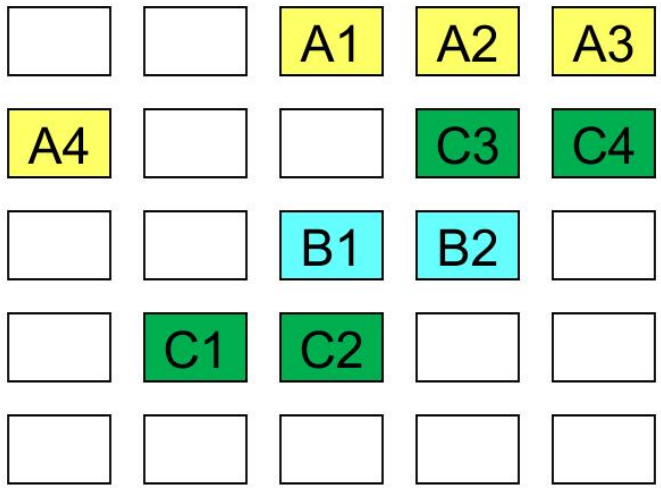
File Allocation

- Each file occupies a number of clusters, not necessarily contiguous
- The file system has to keep track of which clusters are allocated to a file, and which are free



The FAT File System

- Uses a File Allocation Table (FAT) to store the linked list of clusters used by a file



FAT

		3	4	5
nil			9	nil
		13	nil	
	17	8		

Directory \tmp:
File A, 15k, starts at cluster 2
File B, 8k, starts at cluster 12
File C, 13k, starts at cluster 16

FAT16, FAT32

- FAT16, 16 bit is used to address cluster. Max number of cluster = 2^16.
- FAT32: 32 bit is used to address cluster. Max number of cluster = 2^32.
- For a comparison: http://www.ntfs.com/ntfs_vs_fat.htm

Size of FAT

For small disk, e.g. floppy, the FAT is small and so can be kept in memory.

1.44M floppy FAT 16, 512 bytes cluster

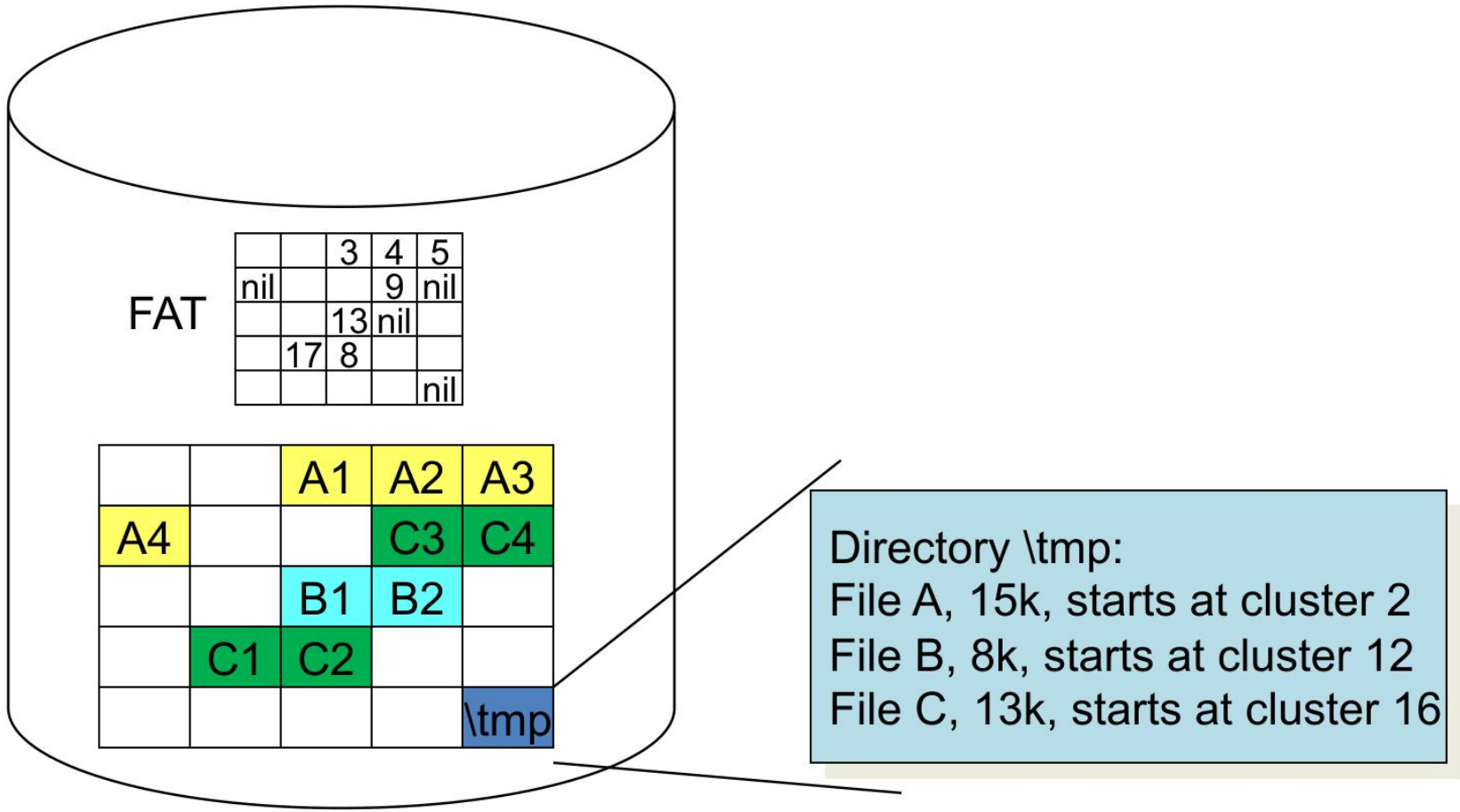
Number of cluster = 1.44M / 512 = 2847
Size of FAT = 16 bit * 2847 = 5694 byte

In a large FAT partition, the File Allocation Table is too large to be cached in memory all the time. File operation on large, highly fragmented file will be slow.

8G FAT 32 partition using 4k cluster

Number of cluster = 8G / 4k = 2 M = 2 * 2^20
Size of FAT = 32 bit * 2 M = 8 MB

FAT File System on Disk



Windows File System

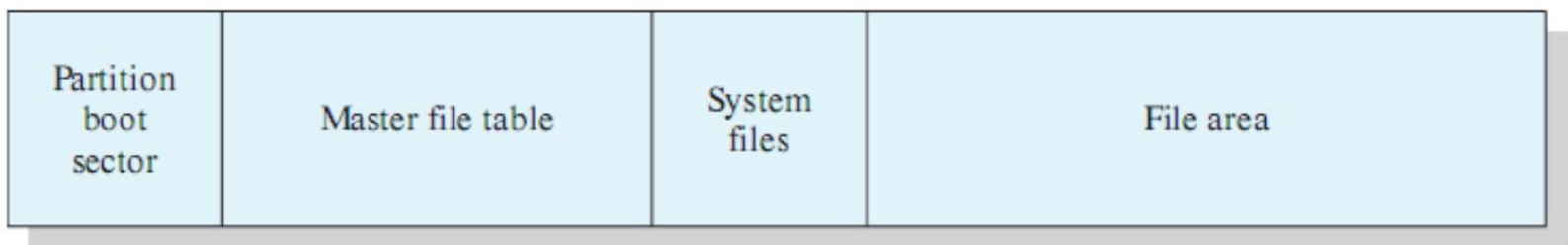
- Key features of New Technology File System (NTFS)
  - Recoverability
  - Security
  - Large disks and large files
  - Multiple data streams
  - Journaling
  - Compression and Encryption

NTFS Volume and File Structure

- Sector
  - The smallest physical storage unit on the disk
  - Almost always 512 bytes
- Cluster
  - One or more contiguous sectors
  - Size in sectors is a power of 2
- Volume
  - Logical partition on a disk, consisting one or more clusters and used by a file system to allocate space

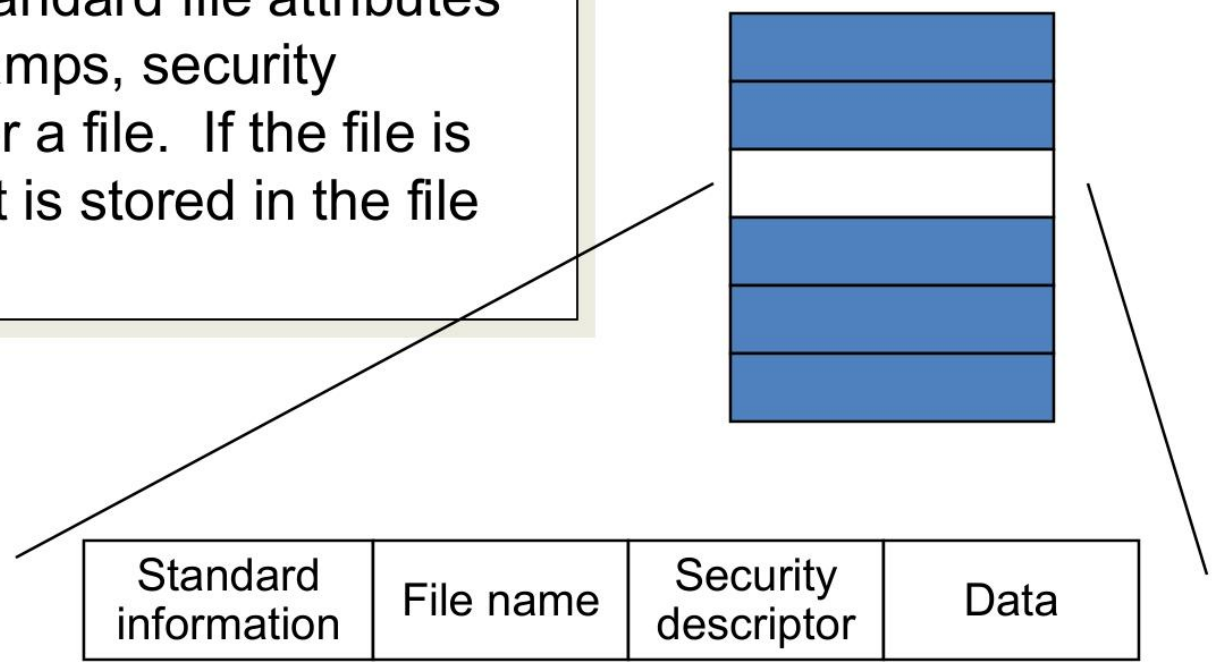
NTFS Volume Layout

- Every element on a volume is a file, and every file consists of a collection of attributes.
  - Even the data contents of a file is treated as an attribute.

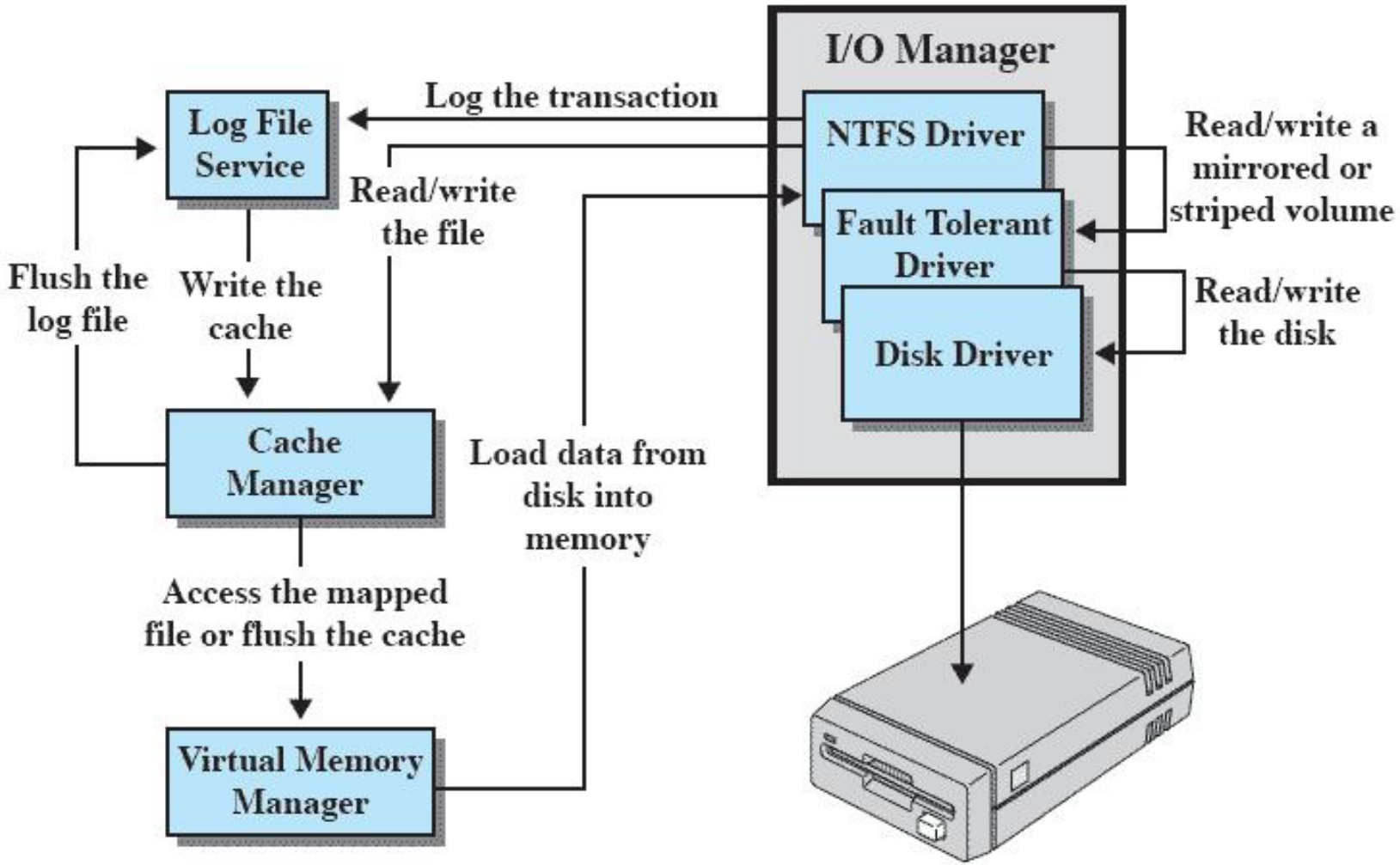


Master File Table (MFT)

In NTFS, (almost) all metadata about the files are stored in a system file name $MFT. Each row of MFT is 1024 bytes long and stores the name(s), standard file attributes like read-only, time stamps, security descriptor, and data for a file. If the file is very small, the content is stored in the file record inside MFT.



Windows NTFS Components



Next Topic

- Exercise
- Revision